# LLE Q&A Report

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## 1 Summary of Symbols

This section presents the definitions and meanings of all relevant symbols related to Locally Linear Embedding (LLE), as discussed in the paper. These symbols encompass key elements, such as variables, vectors, and matrices, required for embedding high-dimensional data into a lower-dimensional space. The symbol table is organized alphabetically and categorized by symbol type for easier reference.

Table 1: Summary of Symbols

| Notation | Definition | Corresponding to |
|---|---|---|
| $d$ | Low-dim embedding dimension | Manifold's intrinsic dimensionality |
| $D$ | High-dim input dimension | Original data space dimensions |
| $E(W)$ | Reconstruction error | Cost function for reconstruction weights |
| $K$ | Number of nearest neighbors | Size of local neighborhood |
| $N$ | Total data points | Total sample size in the dataset |
| $\mathrm{Tr}(\mathbf{G})$ | Trace of $\mathbf{G}$ | Sum of diagonal elements of Gram matrix |
| $W_{ij}$ | Weight element | Contribution of $j$ to $i$ reconstruction |
| $\Phi(Y)$ | Embedding cost | Cost function for low-dimensional embedding |
| $\Delta$ | Regularization parameter | Controls penalty strength |
| $\vec{X}_i$ | Input vector | $i$-th high-dimensional data point |
| $\vec{Y}_i$ | Output vector | $i$-th low-dimensional embedded coordinate |
| $\vec{\eta}_j$ | Neighbor vector | Neighbor of data point $i$ |
| $\mathbf{W}$ | Weight matrix | Local reconstruction weights |
| $\mathbf{G}$ | Gram matrix | Geometric relations among neighbors |
| $\mathbf{M}$ | Cost matrix | Sparse matrix for embedding eigenvalue problem |
| $\mathbf{I}$ | Identity matrix | Unit matrix for covariance constraint |

## 2 Explain Locally in the Context of Locally Linear Embedding in Mathematics

Mathematically, the term *locally* refers to a small neighborhood around a given data point. The data may lie on a curved manifold globally. Within a sufficiently small region, that manifold is approximately flat and can be treated as a Euclidean space. This concept is analogous to the surface of the Earth, which is curved as a whole but appears flat when viewed over a small area.

The term *locally* in LLE means that each data point is reconstructed only from its nearby neighbors on the underlying manifold. This locality is reflected in the sparsity constraint imposed in the objective function. Specifically, for each data point $\vec{X}_i \in \mathbb{R}^D$, the algorithm minimizes the local reconstruction error:

$$E(W) = \sum_{i=1}^{N} \left| \vec{X}_i - \sum_{j \in \mathcal{N}(i)} W_{ij} \vec{X}_j \right|^2, \tag{1}$$

where $\mathcal{N}(i)$ denotes the set of $K$ nearest neighbors of $\vec{X}_i$. $W_{ij}$ represents the linear coefficient, or contribution, of sample $\vec{X}_j$ used to reconstruct $\vec{X}_i$ within its local neighborhood. It encodes the relative geometric relationships between $\vec{X}_i$ and its neighbors. The detailed derivation and solution procedure for $W_{ij}$ is provided in Appendix A. The sparsity constraint in Eq. 1 specifies that $W_{ij}$ is allowed to be nonzero only when $\vec{X}_j$ is a neighbor of $\vec{X}_i$, which can be formally written as

$$W_{ij} = 0 \quad \text{if} \quad \vec{X}_j \notin \mathcal{N}(i). \tag{2}$$

This constraint ensures that each point relies solely on its local neighbors for reconstruction, without referencing distances or relations between distant data points. Therefore, the cost function $E(W)$ reflects the assumption that the manifold is approximately linear within each small neighborhood.

In the subsequent embedding step, the same locality principle is preserved by minimizing

$$\Phi(Y) = \sum_{i=1}^{N} \left| \vec{Y}_i - \sum_{j \in \mathcal{N}(i)} W_{ij} \vec{Y}_j \right|^2, \tag{3}$$

where $\vec{Y}_i \in \mathbb{R}^d$ are the low-dimensional representations of the data. This formulation enforces that the local linear relationships encoded by $W_{ij}$ in the high-dimensional space are preserved in the embedding space. Therefore, LLE preserves the local geometric structure by maintaining the relative relationships among points within each neighborhood. When these local neighborhoods overlap and align consistently across the global space, the overall structure of the data manifold is naturally reconstructed.

# 3 Explain Linear in the Context of Locally Linear Embedding in Mathematics

In mathematics, a transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ is said to be *linear* if it satisfies the two fundamental properties of additivity and homogeneity, *i.e.*,

$$T(a\vec{u} + b\vec{v}) = aT(\vec{u}) + bT(\vec{v}), \tag{4}$$

where $\vec{u}$ and $\vec{v}$ denote arbitrary vectors, while $a$ and $b$ are real-valued scalars. This definition shows that linearity involves only vector addition and scalar multiplication, without powers, products, or other nonlinear operations. Geometrically, many inherently nonlinear structures can be approximated as linear at a sufficiently small scale through a first-order (tangent) approximation. This forms the theoretical basis of the local linearity assumption in LLE.

In LLE, the *linear* refers specifically to the locally linear reconstruction. While the global data distribution lies on a nonlinear manifold, each small neighborhood surrounding a data point can be approximated by a linear subspace. Within each neighborhood, a data sample $\vec{X}_i$ can be represented as a linear reconstruction of its $K$ nearest neighbors:

$$\vec{X}_i \approx \sum_{j \in \mathcal{N}(i)} W_{ij} \vec{X}_j, \tag{5}$$

where $W_{ij}$ denotes the linear reconstruction weight measuring the contribution of neighbor $\vec{X}_j$ to the reconstruction of $\vec{X}_i$. In the subsequent embedding step, LLE aims to find low-dimensional coordinates $\vec{Y}_i \in \mathbb{R}^d$ $(d \ll D)$ that preserve the same local linear relationships by minimizing Eq. 3.

This optimization can be viewed as identifying a linear mapping between the local coordinate frames in the high-dimensional and low-dimensional spaces. In each neighborhood, a linear transformation consisting of translation, rotation, and rescaling maps the high-dimensional local patch to its low-dimensional counterpart. Because the same reconstruction weights $W_{ij}$ are used in both spaces, the relative linear relationships among neighboring points are preserved.

Although the overall transformation from $\mathbf{X}$ to $\mathbf{Y}$ is globally nonlinear, it is constructed from a collection of locally linear mappings that align consistently across neighborhoods. Therefore, LLE can be concisely described as globally nonlinear but locally linear. Its strength lies in combining many locally valid linear approximations to recover the global nonlinear manifold structure.

# 4 Explain Embedding in the Context of Locally Linear Embedding in Mathematics

In mathematics, *embedding* generally refers to a mapping that places one structured space within another while preserving the essential properties of the original space. Formally, an embedding is a one-to-one mapping $f : M \to N$ such that the image $f(M)$ retains the structural characteristics of $M$.

In the framework of LLE, the term *embedding* denotes the process of mapping high-dimensional observations $\vec{X}_i \in \mathbb{R}^D$ onto a low-dimensional coordinate system $\vec{Y}_i \in \mathbb{R}^d$ $(d \ll D)$. This process is achieved by preserving in the low-dimensional space the locally linear reconstruction relationships from the original space. Specifically, LLE determines reconstruction weights $W_{ij}$ that best express each data point as a linear combination of its neighbors. Then seeks low-dimensional coordinates $\vec{Y}_i$ minimizing the embedding cost function

$$\Phi(Y) = \sum_{i=1}^{N} \left| \vec{Y}_i - \sum_{j \in \mathcal{N}(i)} W_{ij} \vec{Y}_j \right|^2 . \tag{6}$$

Minimizing this quadratic objective reduces to solving a sparse eigenvalue problem. Its lowest d+1 nonzero eigenvectors provide the embedding coordinates. The detailed solution procedure is provided in the Appendix B.

Geometrically, the embedding step in LLE constructs a globally nonlinear but locally linear mapping. Each neighborhood undergoes a series of linear transformations, including translation, rotation, and rescaling, which align local patches of the manifold into a coherent global coordinate system. Through the consistent alignment of these overlapping local neighborhoods, LLE recovers the intrinsic low-dimensional structure underlying the high-dimensional data.

# 5 The Assumption of LLE

## 5.1 Manifold Assumption

LLE assumes that high-dimensional data points $\vec{X}_i \in \mathbb{R}^D$ are sampled from a low-dimensional smooth manifold $\mathcal{M} \subset \mathbb{R}^D$. Specifically, there exists a differentiable mapping $f : \mathbb{R}^d \to \mathbb{R}^D$ such that each data point $\vec{X}_i$ can be represented by a low-dimensional coordinate $\vec{Y}_i \in \mathbb{R}^d$ (where $d \ll D$):

$$\vec{X}_i = f(\vec{Y}_i). \tag{7}$$

This assumption implies that, in practical applications, many datasets (such as face images or handwritten digits) are not randomly distributed in high-dimensional space. Instead, they are distributed along a low-dimensional manifold. In short, the core idea behind this assumption is that the data are not randomly scattered in high-dimensional space, but are embedded in, or near, a smooth, low-dimensional, nonlinear manifold. Therefore, this assumption provides LLE with a clear objective: to reveal the intrinsic geometric structure of the data by recovering the low-dimensional coordinates of the manifold.

The mathematical background of the manifold assumption comes from the intrinsic properties of manifolds. In local regions, a manifold can be approximated linearly. The smoothness of the manifold allows its geometric structure to be described using local coordinates. This property enables the recovery of the low-dimensional structure without compromising key geometric features. Without the manifold assumption, the data might not follow any low-dimensional manifold structure. In such a case, the dimensionality reduction process would fail to capture the essential characteristics of the data, thus rendering the reduction process meaningless.

## 5.2 Local Linearity Assumption

LLE assumes that when the manifold is smooth and the data are sampled densely enough, each point and its nearby neighbors approximately lie within a locally low-dimensional linear subspace. Specifically, for each data point $\vec{X}_i \in \mathbb{R}^D$, the other data points $\vec{X}_j$ within its neighborhood $\mathcal{N}(i)$ can be approximately reconstructed by a linear combination, *i.e.*,

$$\vec{X}_i \approx \sum_{j \in \mathcal{N}(i)} W_{ij} \vec{X}_j, \tag{8}$$

where $W_{ij}$ is the reconstruction weight between the data points $\vec{X}_i$ and $\vec{X}_j$, and

$$\sum_{j \in \mathcal{N}(i)} W_{ij} = 1. \tag{9}$$

This assumption is a direct consequence of the smoothness of the manifold. Mathematically, the smoothness of the manifold ensures that within the neighborhood of each point, the data variation is linear and can be described by the tangent space. While the manifold may be nonlinear globally, within the neighborhood of each data point, the variation of the manifold can be approximated by a low-dimensional linear space. This enables LLE to reconstruct data points through linear combinations, thereby simplifying the dimensionality reduction process.

The validity of the local linear assumption depends on the smoothness of the manifold and the density of the sampling. If the neighborhood is too large, the linear approximation of the manifold will no longer hold. Conversely, if the neighborhood is too small, it may fail to capture the geometric structure of the data accurately. Therefore, the choice of neighborhood size is crucial for LLE. Typically, the size of the neighborhood $K$ satisfies $d < K \ll N$. This means that the neighborhood should not be too small or too large to ensure the validity of the linear approximation.

### 5.3 Local Geometry Preservation Assumption

LLE assumes that in the low-dimensional space, the low-dimensional representation of each data point $\vec{Y}_i \in \mathbb{R}^d$ should preserve the geometric structure of its high-dimensional neighborhood. Specifically, the low-dimensional representation should be able to reconstruct its neighboring points $\vec{Y}_j$ using the same reconstruction weights $W_{ij}$, *i.e.*,

$$\vec{Y}_i \approx \sum_{j \in \mathcal{N}(i)} W_{ij} \vec{Y}_j. \tag{10}$$

This is achieved by minimizing the cost function:

$$\Phi(Y) = \sum_{i=1}^{N} \left| \vec{Y}_i - \sum_{j \in \mathcal{N}(i)} W_{ij} \vec{Y}_j \right|^2. \tag{11}$$

The essence of this assumption is that LLE prioritizes the preservation of local geometric properties during the embedding process, rather than focusing solely on global distance information. By minimizing the cost function, LLE links the low-dimensional embedding with the high-dimensional reconstruction weights $W_{ij}$. This ensures that the local geometric information in high-dimensional space is faithfully transferred to the low-dimensional space. Consequently, the data points in the low-dimensional space can maintain the local structure of their high-dimensional counterparts.

This assumption is fundamental to LLE. It enables LLE to preserve the local geometric features of the data during nonlinear dimensionality reduction, minimizing distortions and information loss in the reduction process.

## 6 The Problems to be Solved by LLE

LLE aims to address the problem of nonlinear dimensionality reduction for high-dimensional data while preserving the local geometric structure of the data. Specifically, for $N$ high-dimensional data points $X = \{\vec{X}_1, \vec{X}_2, \ldots, \vec{X}_N\}$, where $\vec{X}_i \in \mathbb{R}^D$, the goal of LLE is to find their embedding in a low-dimensional space $Y = \{\vec{Y}_1, \vec{Y}_2, \ldots, \vec{Y}_N\}$, where $\vec{Y}_i \in \mathbb{R}^d$ $(d \ll D)$. The embedding should preserve the local geometric structure of the original high-dimensional data as much as possible. It should also avoid the severe distortions often introduced by linear methods such as Principal Component Analysis (PCA) or Multidimensional Scaling (MDS).

The core of this problem lies in the fact that the data are often distributed on a nonlinear manifold. As a result, the Euclidean distance in high-dimensional space cannot accurately reflect the true neighborhood relationships of the data points on the manifold. Linear dimensionality reduction algorithms, such as PCA or MDS, typically attempt to find a global optimal linear projection or preserve global Euclidean distances. However, these approaches often result in incorrectly mapping points that are far apart in high-dimensional space to nearby locations in the low-dimensional space. This typically results in severe distortions of the data's local geometry. For example, when an S-shaped manifold is reduced using PCA, points located on different curved regions may be projected onto similar positions. This causes them to be squeezed together, leading to the loss of the original topological structure of the data (as shown in Fig. 1).

LLE provides an efficient solution to this problem, with its core based on the local linearity assumption. Specifically, LLE assumes that any nonlinear manifold can be approximated as linear within a sufficiently small neighborhood. Based on this, the LLE algorithm achieves dimensionality reduction by solving two closely related mathematical optimization problems.

First, LLE captures the local geometric structure of each data point in high-dimensional space. Specifically, it assumes that each data point $\vec{X}_i$ can be linearly reconstructed by its $K$ nearest neighbors $\mathcal{N}_i = \{X_j\}_{j=1}^{K}$. To find the optimal reconstruction weights $W_{ij}$, LLE minimizes the reconstruction error in high-dimensional space:

$$E(W) = \sum_{i=1}^{N} |\vec{X}_i - \sum_{j \in \mathcal{N}_i} W_{ij} \vec{X}_j|^2. \tag{12}$$

This optimization problem involves two crucial constraints:

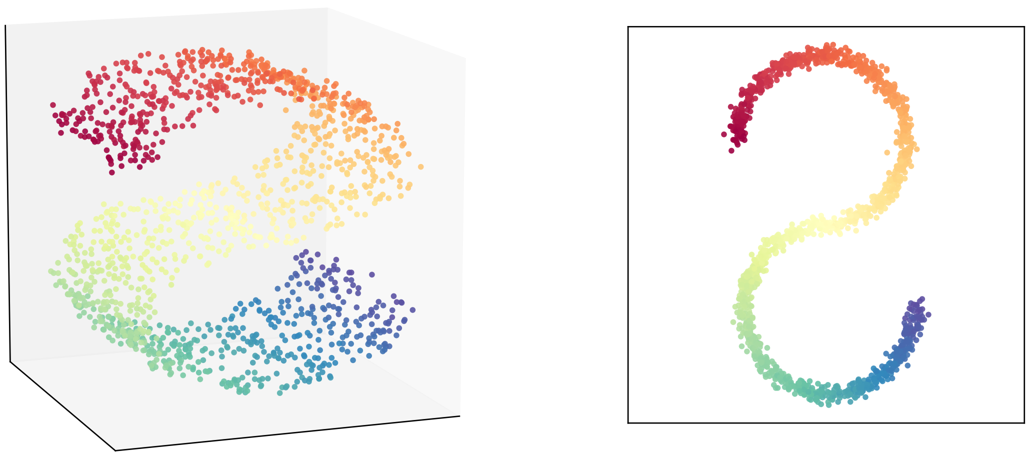


Figure 1: Original data vs. Data after PCA dimensionality reduction

- **Sparsity constraint:** LLE requires that the reconstruction of each data point depends only on its neighbors. This means that only the neighbors $\vec{X}_j$ of point $\vec{X}_i$ contribute to its reconstruction, while for points that are not within the neighborhood, the reconstruction weight is zero. This can be formally expressed as:
$$\mathrm{W}_{ij} = \begin{cases} 1, \vec{X}_j \in \mathcal{N}_i \\ 0, \vec{X}_j \notin \mathcal{N}_i. \end{cases} \tag{13}$$
The introduction of this constraint ensures that the reconstruction of each data point depends only on its local neighborhood. It enables LLE to capture the local geometric structure of the data accurately. This also guarantees that the dimensionality reduction process is not influenced by data points that are distant from the current point.

- **Invariance constraint:** LLE requires that the sum of the reconstruction weights for each data point must equal 1, *i.e.*,
$$\sum_{j \in \mathcal{N}_i} W_{ij} = 1. \tag{14}$$
This constraint ensures the insensitivity of local reconfiguration to translation, rotation, and scaling transformations. Specifically, these transformations do not alter the relative positional relationships between data points. Through this constraint, LLE preserves the relative positions within each local neighborhood of the manifold. As a result, it avoids errors that may arise from changes in the coordinate system. A more detailed mathematical proof of this constraint will be provided in Appendix C.

Next, the core idea of LLE is that if these local reconstruction weights $W_{ij}$ faithfully describe the local geometric structure in high-dimensional space, they should remain invariant in the low-dimensional embedding space as well. Therefore, LLE further seeks low-dimensional embedding coordinates $\vec{Y}_i$ such that they can also be linearly reconstructed using the same weights $W_{ij}$ in the low-dimensional space, while minimizing the low-dimensional reconstruction error:

$$\Phi(Y) = \sum_{i=1}^{N} |\vec{Y}_i - \sum_{j \in \mathcal{N}_i} W_{ij}\vec{Y}_j|^2. \tag{15}$$

To ensure that the low-dimensional embedding problem has a unique and meaningful solution, LLE introduces two additional constraints:

- **Translation degree-of-freedom constraint:** LLE requires that the mean value of all embedding points $\vec{Y}_i$ be 0:
$$\sum_{i=1}^{N} Y_i = 0. \tag{16}$$
Through this constraint, LLE removes the translational degree of freedom, ensuring that the geometric structure in the low-dimensional space is unaffected by translation. We provide a more detailed explanation of this constraint in Appendix D.

- **Scale and rotation degree-of-freedom constraint:** LLE requires the covariance matrix of the low-dimensional embedding to be the identity matrix:

$$\frac{1}{N}\sum_{i=1}^{N} Y_i Y_i^T = I. \tag{17}$$

  This constraint fixes the scale and orientation of the low-dimensional embedding, preventing scaling and rotation transformations from affecting the geometric structure of the data. As a result, the low-dimensional representation stably reflects the intrinsic structure of the data. A more detailed explanation of this constraint can be found in Appendix E.

LLE ultimately transforms the local linear reconstruction in high-dimensional space into a global eigenvalue problem in the low-dimensional space. Specifically, by solving a sparse eigenvalue problem, LLE can obtain the low-dimensional embedding coordinates. The core of this process is to construct the low-dimensional embedding based on local geometric properties through a global eigenvalue decomposition. In this way, LLE effectively preserves the intrinsic structure of the data.

# 7 The Limitations of LLE

The LLE algorithm has significant advantages in nonlinear dimensionality reduction. However, there are also limitations closely related to its underlying assumptions, computational process, and dependence on the data. The following is a detailed analysis of these limitations.

## 7.1 Strict Requirements on Data Sampling

LLE assumes that the data points are from a smooth and sufficiently sampled low-dimensional manifold. This implies that the data must be uniformly distributed across different parts of the manifold, and there should be enough samples in local regions to ensure accurate reconstruction of the geometric structure. If the data points are unevenly distributed on the manifold, some local areas may lack sufficient data points to describe their geometric structure accurately. This can lead to increased reconstruction errors and negatively impact the results of dimensionality reduction. When data is sparse in local regions of the manifold, LLE may fail to accurately reconstruct the data through neighbor relationships, resulting in distortion in the embedding. For example, if we sample from a spiral manifold but have sufficient samples only in the middle portion, with sparse data at both ends. In that case, LLE may fail to capture the geometric features at the ends of the spiral, ultimately resulting in a distorted dimensionality reduction of the manifold, as shown in Fig. 2.

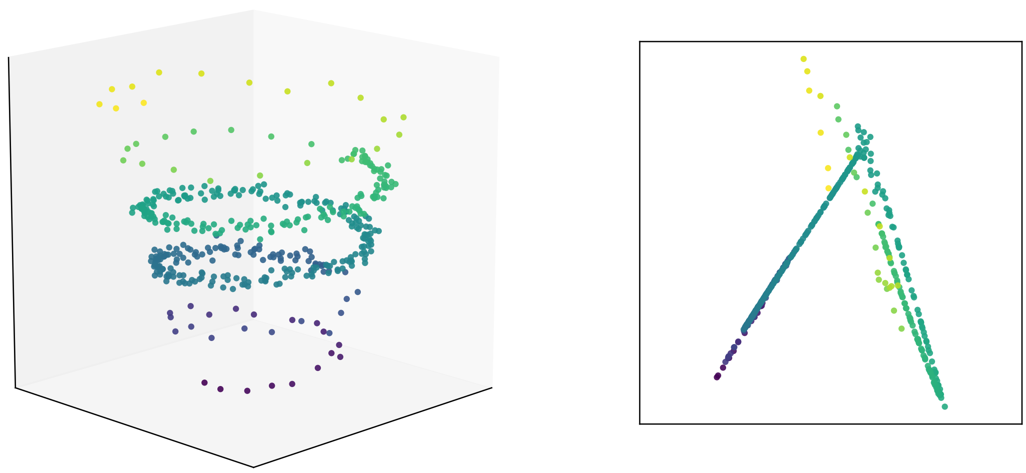
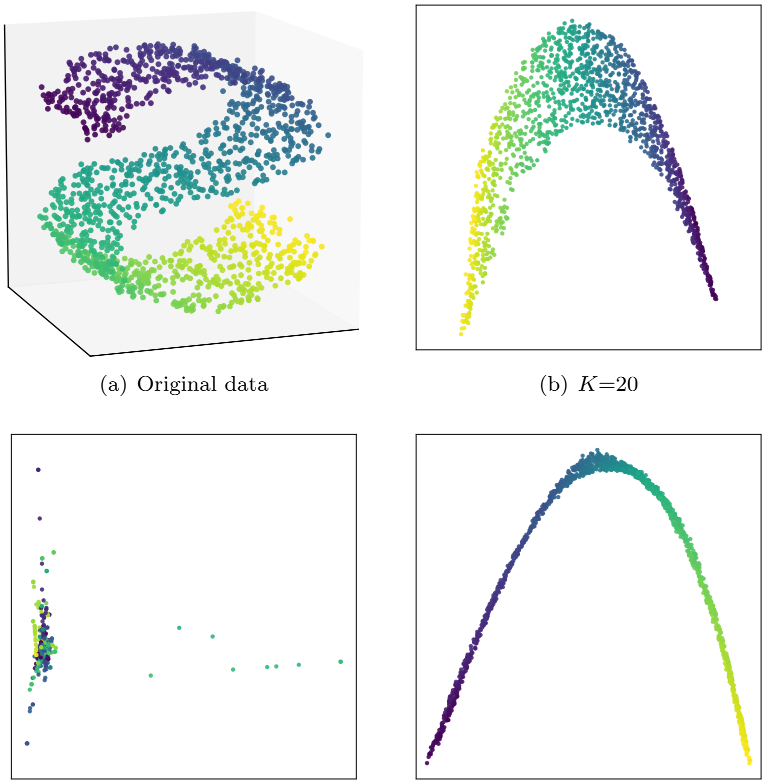


Figure 2: Distortion in dimensionality reduction caused by uneven sampling.

## 7.2 Sensitivity to Nearest Neighbor Selection

LLE performs dimensionality reduction by computing the reconstruction relationships between each data point and its $K$ nearest neighbors. If the number of neighbors $K$ is too small, the local geometric structure may not be effectively captured, leading to increased reconstruction errors. On the other hand, if $K$ is too large, the neighborhood relationships may become too broad, affecting the precise description of local geometric features and causing the loss of manifold details, as shown in Fig. 3. Therefore, selecting an appropriate value for $K$ is crucial for LLE. Both too large and too small values of $K$ can impact the final dimensionality reduction results.



(a) Original data

(b) $K$=20

(c) $K$=3

(d) $K$=60

Figure 3: The impact of different $K$ values on dimensionality reduction results.

## 7.3 Limitations on the Global Topology of the Manifold

LLE focuses only on the relationships within local neighborhoods, optimizing by minimizing local reconstruction errors, without considering the global distances between data points. When global structure is crucial, LLE's local optimization may not preserve the geometric relationships between distant points. Consequently, the embedding can lose the manifold's global topology and misrepresent its geometric properties. For example, when dealing with "barbell" shaped data, LLE often struggles to unfold the "bar" connecting the two "balls." As shown in Fig. 4, it is evident that the dimensionality-reduced result fails to preserve the overall structure.

## 7.4 Limited Ability to Handle Shear Transformations

Although LLE can preserve the geometric invariance under translation, rotation, and scaling transformations, it cannot handle shear transformations. This is because LLE relies on the local linearity assumption, which assumes that data points are reconstructed through linear transformations in local regions. Since shear transformations are nonlinear, they alter the relative angles and distances between data points, thereby changing the local geometric relationships. As LLE is unable to capture such changes, it results in the misplacement of data points in the low-dimensional space. For example, in facial image datasets, the face may undergo local shear transformations under different viewpoints, such as stretching or compressing of the mouth and eyes. LLE cannot accurately handle these transformations, resulting in dimensionality-reduced facial images that fail to reflect the original facial features accurately.

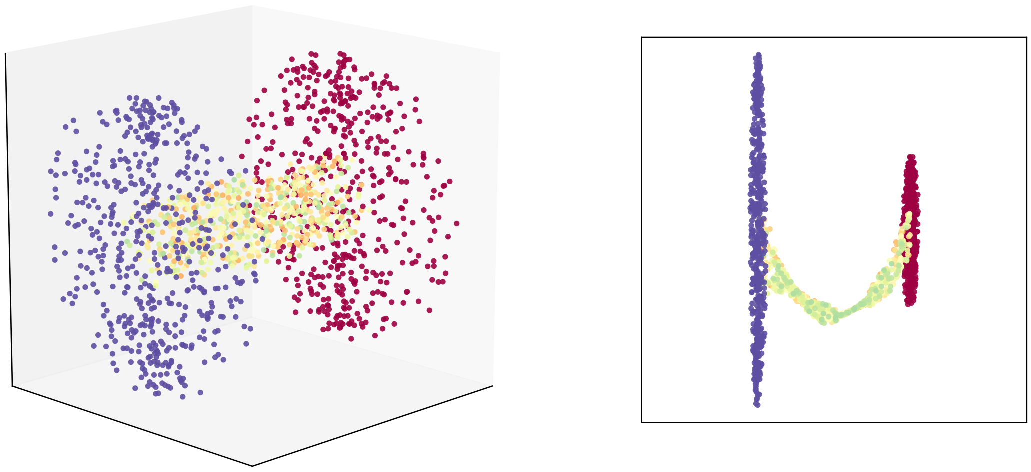


Figure 4: Comparison of the original "barbell" shaped data and the LLE dimensionality reduction result($K$=15)

### 7.5 Unreliability in estimating intrinsic dimensionality

LLE requires the user to predefine the target dimensionality. If the chosen target dimension does not match the intrinsic dimensionality of the data manifold, the resulting embedding may not accurately reflect the manifold's geometric structure. For example, if the manifold is actually two-dimensional but we mistakenly set its dimensionality to three, LLE may introduce an unnecessary extra dimension during the reduction process. This can result in extraneous noise or false redundant dimensions, causing the reduced data to fail to accurately represent the original manifold structure.

### 7.6 Computational Complexity

In the dimensionality reduction process, LLE needs to solve a sparse $N \times N$ eigenvalue problem. The computational complexity of solving the eigenvalue problem increases exponentially with the number of data points. When the dataset is vast, the computational cost of LLE rises sharply, limiting its application to large-scale datasets. Although sparse matrix optimizations can reduce the computational load, for extremely large datasets, LLE still requires significant computational resources and time.

## 8 The difference between PCA and LLE

### 8.1 Core Objectives

- **PCA:** The core goal of PCA is to map data from high-dimensional space to low-dimensional space through linear transformation, while preserving as much of the variation information as possible. By performing eigenvalue decomposition on the data's covariance matrix, PCA identifies the principal components and projects the data onto them. This process simplifies the data structure, removes redundancy and noise, and retains the most important information. Specifically, PCA reduces dimensionality by maximizing the variance of the data in the low-dimensional space, aiming to find the most important directions and reduce redundant information. The objective function for this process is:

$$\underset{\mathbf{P}}{\text{maximize}} \quad \mathbf{P}^T \mathbf{C}_X \mathbf{P}, \tag{18}$$

  where $\mathbf{P}$ is the matrix of principal components. $\mathbf{C}_\mathbf{X}$ represents the covariance matrix of the original data.

- **LLE:** The core goal of LLE is to map data from high-dimensional space to low-dimensional space while preserving the local geometric structure of the data. LLE assumes that the data points are distributed on a low-dimensional manifold and that the neighborhoods of these data points have linear relationships. Therefore, LLE seeks to find the low-dimensional representation of each data point by minimizing its reconstruction error, while retaining the local geometric structure of the data. Specifically, LLE first determines the reconstruction weights by minimizing the reconstruction error:

$$E(W) = \sum_{i=1}^{N} |\vec{X}_i - \sum_{j \in \mathcal{N}_i} W_{ij} \vec{X}_j|^2. \tag{19}$$

Then, LLE optimizes the coordinates in the low-dimensional space by fixing these reconstruction weights. This ensures that the relationships between neighboring points in the low-dimensional embedding are preserved as much as possible:

$$\min_{Y} \sum_{i=1}^{N} |\vec{Y}_i - \sum_{j \in \mathcal{N}_i} W_{ij} \vec{Y}_j|^2. \tag{20}$$

In this way, LLE can project the data into a low-dimensional space while preserving the local structure.

### 8.2 Key Assumptions

- **PCA:** PCA emphasizes global geometric structure by finding directions of maximum variance that capture the main features of the data. It assumes that a global linear transformation can map the data to a lower-dimensional space. The method further assumes that the dominant modes of variation are uniform, so the most significant directions are consistent across the dataset.
- **LLE:** LLE assumes that the data is sampled from a low-dimensional manifold, and that the local regions of the data can be reconstructed using linear methods. LLE focuses on the local geometric structure of the data, rather than global linear relationships. It achieves dimensionality reduction by preserving the linear reconstruction relationships within the local neighborhoods of the data points.

### 8.3 Parameter Settings

PCA is a non-parametric method that requires no parameter settings, with the only decision being the number of principal components to retain after dimensionality reduction. In contrast, the number of nearest neighbors $K$ in LLE is a crucial free parameter. The choice of $K$ directly impacts the results of dimensionality reduction. If $K$ is too small, it will fail to capture sufficient local information, leading to a dimensionality reduction result that does not accurately reflect the data structure. If $K$ is too large, the neighborhood relationships may become too broad, causing the loss of local geometric structure. Therefore, compared to the simple, non-parametric setup of PCA, LLE requires careful selection of an appropriate $K$ value to ensure the accuracy and effectiveness of the dimensionality reduction results.

### 8.4 Pros and Cons

- **PCA:**The main advantage of PCA lies in its simplicity and efficiency, making it particularly well-suited for handling large-scale linear datasets. Since the algorithm is based on eigenvalue decomposition, the computational process is relatively straightforward. This makes it ideal for tasks that require dimensionality reduction while preserving the data's global structure. Additionally, the principal components of PCA provide strong interpretability, helping to understand the main features of the data. However, because PCA is a linear method, it is not effective at handling nonlinear data. In particular, when the data is distributed on complex manifolds, PCA often fails to capture the nonlinear relationships in the data, leading to poor performance in such cases.
- **LLE:** The advantage of LLE lies in its ability to effectively handle nonlinear data, particularly in cases where the data is distributed on a low-dimensional manifold. By preserving the local geometric structure of the data, LLE can better represent the complex, nonlinear patterns within the data. However, it is sensitive to the choice of parameter $K$, and an improper selection may lead to distorted dimensionality reduction results.

### 8.5 Typical Applications

PCA is commonly used for tasks such as data compression, denoising, and feature extraction. It reduces the dimensionality of the data while retaining its main features and removing redundant information. It is particularly effective when the data structure is nearly linear. In contrast, LLE is mainly used to handle nonlinear data. It is often applied in tasks such as face recognition and speech recognition, where preserving local structure helps capture complex facial expressions and speech patterns more effectively.

Overall, PCA is suited for tasks with simple structures and clear linear relationships, while LLE is better for handling data with complex nonlinear relationships.

# 9 Experiment

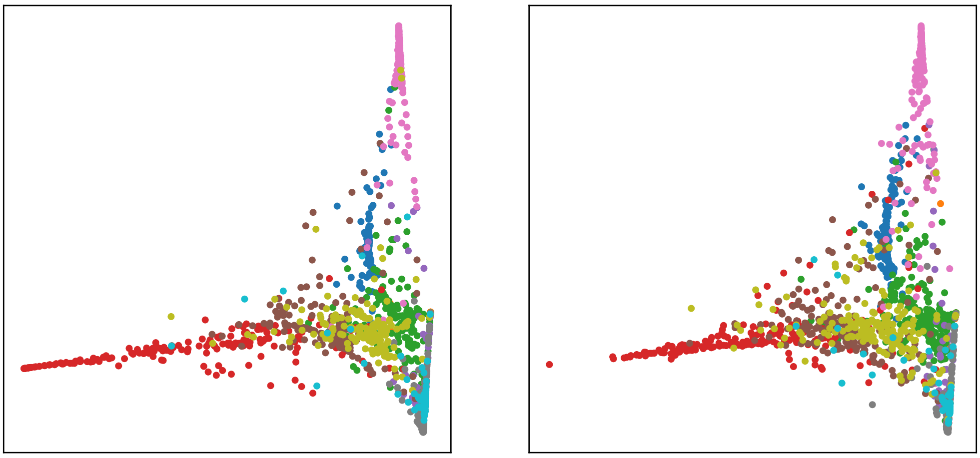
## 9.1 Datasets

The MNIST[1] dataset contains 70,000 grayscale images of handwritten digits ranging from 0 to 9, each with a resolution of $28 \times 28$ pixels. In this experiment, a small subset of the dataset is used, consisting of the first 2,000 training images and the first 2,000 testing images.

## 9.2 Experimental Results and Analysis

We applied the LLE method to perform dimensionality reduction on both the training and test sets. The number of neighbors $K$ is set to 5, and the data are reduced to a two-dimensional space. The visualization results are illustrated in Fig. 5. From the results, we draw the following key observations:

- The embedding of the test set is highly consistent with that of the training set. The spatial positions and topological structures of the main clusters remain stable. This indicates that the low-dimensional manifold learned from the training data generalizes well to unknown samples and maintains good topological stability. The results also confirm that the chosen parameter, $K = 5$, is appropriate and ensures the stable reconstruction of local geometry.
- After LLE maps the high-dimensional features into two dimensions, the samples within each class show clear clustering. Samples of the same class are continuously distributed along the low-dimensional manifold. This indicates that LLE effectively preserves local neighborhood relationships and reveals the global nonlinear structure of the data.
- There remains considerable overlap between classes, indicating that LLE primarily emphasizes local geometric relationships rather than class separability. As a result, it shows limited effectiveness in achieving global separation between different classes.



(a) Train set visualization

(b) Test set visualization

Figure 5: Visualization results of dimensionality reduction

## 9.3 Parameter Analysis

To investigate the influence of parameters on classification performance, we conducted a parameter analysis on the number of neighbors $K$ and the embedding dimension $d$. Specifically, we set $K, d \in \{2, 5, 10, 15, 20, 25, 30, 40, 50, 60\}$. For each $(K, d)$ pair, we first obtain the low-dimensional representation using LLE. We then train a 1-nearest neighbor (1-NN) classifier on this representation and evaluate its accuracy on the test set. The classification accuracies corresponding to all $(K, d)$ combinations were visualized as a heatmap, as shown in Fig. 6. The key findings are summarized as follows.

[1] http://www.cad.zju.edu.cn/home/dengcai/Data/MLData.html

- When $K$ is small, the accuracy rate is at a relatively low level. This indicates that an overly small neighborhood cannot reliably capture local structure. As $K$ increased from 5 to 15, the accuracy rate gradually improved and stabilized. When $K$ continues to increase, the performance starts to decline. This shows that an excessively large neighborhood breaks the locality assumption, increases reconstruction error, and reduces the discriminative ability of the embedding.

- In the lower-dimensional range $d = 2 \sim 15$, accuracy increases steadily with $d$. This indicates that higher embedding dimensions capture more discriminative information on the manifold. When $d > 20$, the improvement saturates and the gains are limited. This suggests that the principal geometric information is already expressed within a moderate dimensional range.

- When $K$ is between 5 and 10 and $d$ is within the range of $20 \sim 40$, the model achieves superior and stable classification performance. This result indicates that LLE exhibits a certain degree of robustness to parameter variations. With an appropriate neighborhood size and embedding dimension, the model can reliably extract low-dimensional representations that reflect the nonlinear structure of the data.

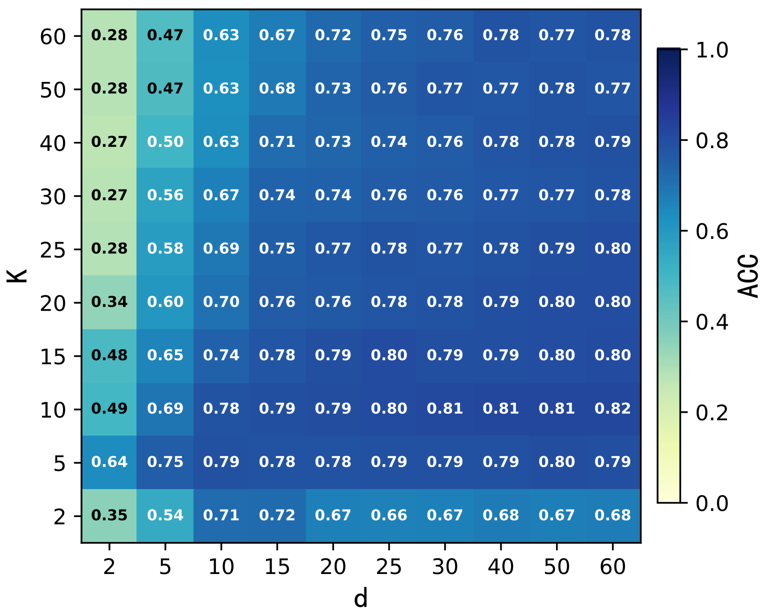
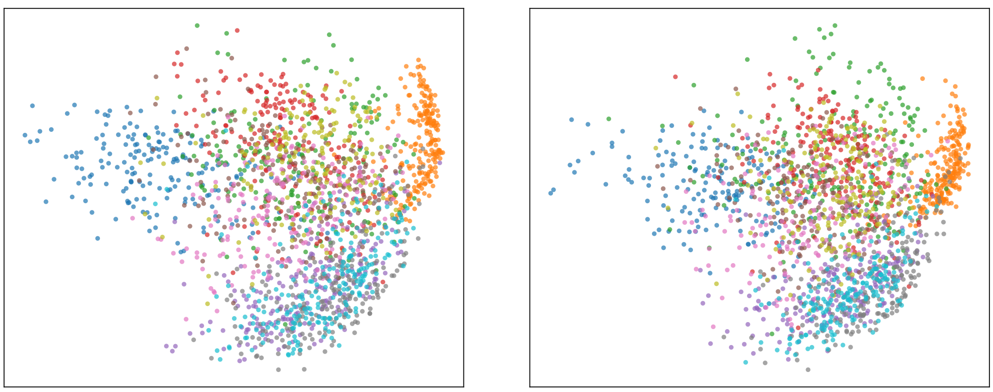



Figure 6: Heatmap of Classification Accuracy over LLE Parameters ($K$ and $d$)



(a) Train set visualization

(b) Test set visualization

Figure 7: Dimensionality reduction results using PCA

### 9.4 Contrast Experiment

To more comprehensively evaluate the effectiveness of LLE, we compare it with the commonly used linear dimensionality reduction method PCA in terms of classification performance and two-dimensional visualization. The visualization results of PCA are shown in Fig. 7, and the classification comparison is presented in Fig. 8. From the experiments, we observe:

- After dimensionality reduction with PCA (see Fig. 7), samples from different classes are scattered on the two-dimensional plane. Inter-class overlap is pronounced, and clear clusters are hard to obtain clear clusters. This suggests that PCA struggles to preserve manifold topology in low-dimensional space. In contrast, the LLE visualization (see Fig. 5) shows a stronger clustering tendency. Same-class samples are more compact, and inter-class separation is more evident. As a result, LLE better reveals the underlying nonlinear manifold structure.
- Fig. 8 shows that LLE performs better in the low-dimensional regime. As the embedding dimension increases, the accuracies of both PCA and LLE rise and then stabilizes. When the dimension reaches 10, PCA begins to surpass LLE. This indicates that LLE is more effective at revealing local nonlinear structure in low-dimensional spaces, whereas PCA recovers global variance more effectively at higher dimensions. As the dimension grows further, the local advantage of LLE weakens, and PCA's strengths in linear reconstruction and variance preservation become dominant.

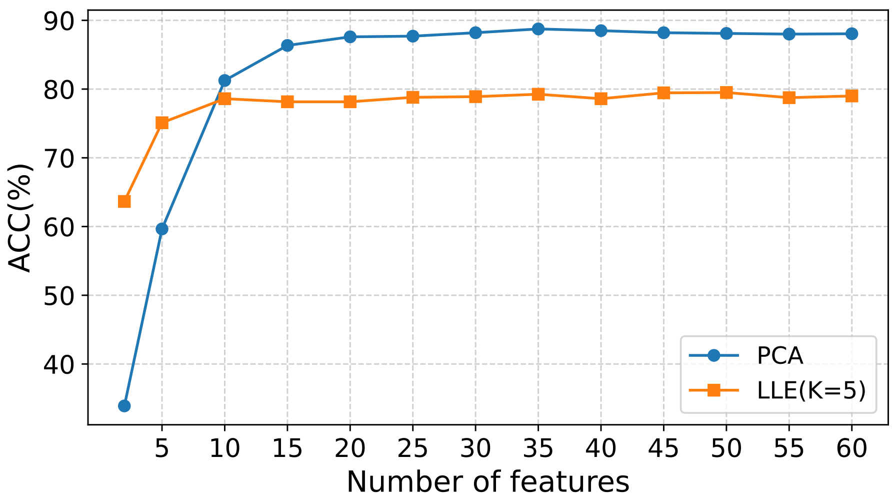



Figure 8: Comparison of classification accuracy between PCA and LLE under different numbers of embedding features.

# Appendix

## A Solution of Linear Reconstruction Weights

In this section, we provide a detailed derivation of the reconstruction weights $W_{ij}$ using the method of Lagrange multipliers.

First, LLE defines the reconstruction error for a single data point $\vec{x}$:

$$\varepsilon = \left|\vec{x} - \Sigma_j w_j \vec{\eta}_j\right|^2, \tag{21}$$

where $\vec{\eta}_j$ is the $j$-th neighbor of $\vec{x}$. $W_j$ denotes the reconstruction weight of $\vec{\eta}_j$ for $\vec{x}$. To ensure local geometric invariance under translation, rotation, and scaling, LLE imposes the following constraints:

$$\sum_j W_j = 1. \tag{22}$$

Combining Eq. 21 and Eq. 22, the reconstruction error can be written as:

$$\begin{aligned}
\varepsilon &= \left|\sum_j W_j \left(\vec{x} - \overrightarrow{\eta_j}\right)\right|^2 \\
&= \left(\sum_j W_j \left(\vec{x} - \overrightarrow{\eta_j}\right)\right) \cdot \left(\sum_{\mathrm{k}} W_k \left(\vec{x} - \overrightarrow{\eta_k}\right)\right) \\
&= \sum_j \sum_k W_j W_k \left(\vec{x} - \overrightarrow{\eta_j}\right)\left(\vec{x} - \overrightarrow{\eta_k}\right).
\end{aligned} \tag{23}$$

To further simplify the expression, LLE introduces the local Gram matrix $G$. Its elements $G_{jk}$ are defined as:

$$G_{jk} = \left(\vec{x} - \vec{\eta}_j\right) \cdot \left(\vec{x} - \vec{\eta}_k\right). \tag{24}$$

The $K \times K$ matrix captures the relative geometry between the center point $\vec{x}$ and its neighbors. Substituting Eq. 24 into Eq. 23 yields a concise expression for the reconstruction error:

$$\varepsilon = \Sigma_{jk} W_j W_k G_{jk}. \tag{25}$$

Our goal is to minimize $\varepsilon$ subject to the constraint in Eq. 22. This constrained optimization problem can be solved using the method of Lagrange multipliers. The Lagrange function can be formulated as:

$$\mathcal{L}\left(W_j, \lambda\right) = \sum_{jk} W_j W_k G_{jk} - \lambda \left(\sum_j W_j - 1\right). \tag{26}$$

To find the minimum value, we take the partial derivative of the Lagrangian with respect to $W_j$ and set it to 0, that is,

$$\frac{\partial \mathcal{L}}{\partial W_j} = \frac{\partial}{\partial W_j}\left(\sum_{jk} W_j W_k G_{jk}\right) - \frac{\partial}{\partial W_j}\left(\lambda \sum_j W_j - \lambda\right) = 0. \tag{27}$$

Since $G_{jk} = G_{kj}$, the above equation can be rewritten as:

$$\frac{\partial \mathcal{L}}{\partial W_j} = 2 \sum_k W_k G_{jk} - \lambda = 0. \tag{28}$$

It then follows that:

$$W_j = \frac{\lambda}{2} \Sigma_k G_{jk}^{-1}. \tag{29}$$

By combining Eq. 22 and Eq. 29, we can derive that:

$$\frac{\lambda}{2} \Sigma_j \Sigma_k G_{jk}^{-1} = 1, \tag{30}$$

$$\frac{\lambda}{2} = \frac{1}{\Sigma_j \Sigma_k G_{jk}^{-1}} = \frac{1}{\Sigma_{lm} G_{lm}^{-1}}. \tag{31}$$

Substituting Eq. 31 into Eq. 29 yields the final solution:

$$W_j = \frac{\Sigma_k G_{jk}^{-1}}{\Sigma_{lm} G_{lm}^{-1}}. \tag{32}$$

# B Solution of Low-dimensional Embedding Coordinates

In this section, building on the previously obtained reconstruction weights $W_{ij}$, we provide a detailed derivation of the solution process for the low-dimensional embedding coordinates.

We start from the cost function of the embedding stage:

$$\Phi(Y) = \sum_i |\vec{Y}_i - \sum_j W_{ij}\vec{Y}_j|^2. \tag{33}$$

For notational convenience, LLE introduces the Kronecker $\delta$, which is formally defined as:

$$\delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j. \end{cases} \tag{34}$$

We can rewrite Eq. 33 as:

$$\begin{aligned} \Phi(Y) &= \sum_i |\sum_j (\delta_{ij} - W_{ij})\vec{Y}_j|^2 \\ &= \sum_i \left(\sum_j (\delta_{ij} - W_{ij})\vec{Y}_j\right)^T \left(\sum_k (\delta_{ik} - W_{ik})\vec{Y}_k\right) \\ &= \sum_i \sum_j \sum_k (\delta_{ij} - W_{ij})(\delta_{ik} - W_{ik})\vec{Y}_j^T\vec{Y}_k \\ &= \sum_{jk} M_{jk}\vec{Y}_j^T\vec{Y}_k, \end{aligned} \tag{35}$$

where $M_{jk} = \sum_i (\delta_{ij} - W_{ij})(\delta_{ik} - W_{ik})$, which can also be expanded as:

$$M_{jk} = \delta_{jk} - W_{jk} - W_{kj} + \sum_i W_{ij}W_{ik}. \tag{36}$$

In the following, for notation convenience, we stack the low-dimensional coordinates of all samples by rows into a matrix:

$$Y = \begin{bmatrix} \vec{Y}_1^T \\ \vec{Y}_2^T \\ \vdots \\ \vec{Y}_N^T \end{bmatrix} \in \mathbb{R}^{N\times d}. \tag{37}$$

Based on this, we return to Eq. 35 and obtain that:

$$\begin{aligned} \Phi(Y) &= \sum_{jk} M_{jk}(YY^T)_{kj} \\ &= \sum_{jk} M_{jk} \sum_{p=1}^{d} Y_{kp}Y_{jp} \\ &= \sum_{p=1}^{d} \sum_{jk} M_{jk}Y_{jp}Y_{kp} \\ &= \sum_{p=1}^{d} (Y^TMY)_{pp} \\ &= \mathrm{Tr}(Y^TMY). \end{aligned} \tag{38}$$

This optimization problem is subject to the unit covariance constraint:

$$Y^TY = NI, \tag{39}$$

where $I$ is the $d\times d$ identity matrix. Therefore, we introduce a Lagrange multiplier matrix $\Lambda$ and construct a Lagrangian function to solve this problem. The Lagrangian $\mathcal{L}$ can be defined as:

$$\mathcal{L}(Y, \Lambda) = \mathrm{Tr}\left(Y^TMY\right) - \mathrm{Tr}\left(\Lambda\left(Y^TY - NI\right)\right). \tag{40}$$

To find the minimum, we take the derivative of $\mathcal{L}$ with respect to $Y$ and set it to 0. We then obtain:

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial Y} &= \frac{\partial \operatorname{Tr}\left(Y^T M Y\right)}{\partial Y} - \frac{\partial \operatorname{Tr}\left(\Lambda Y^T Y\right)}{\partial Y} \\ &= 2MY - 2Y\Lambda \\ &= 0.\end{aligned} \tag{41}$$

From this, we obtain $MY = Y\Lambda$, which takes the form of a generalized eigenvalue problem. Therefore, the optimal solution is given by the smallest nonzero eigenvectors of $M$.

# C Affine Invariance Constraint Proof

The invariance constraints in the LLE algorithm require that the sum of the reconstruction weights for each data point must equal 1, *i.e.*,

$$\sum_{j \in \mathcal{N}_i} W_{ij} = 1. \tag{42}$$

The purpose of this constraint is to ensure that the reconstruction weights $W_{ij}$ remain invariant under affine transformations such as translation, rotation, and scaling. To demonstrate this, we derive the effects of translation, rotation, and scaling. This shows why the invariance constraint makes LLE's reconstruction insensitive to these affine transformations.

## C.1 Translation Invariance

Translation is a special case of an affine transformation, represented as:

$$\vec{X}_i' = \vec{X}_i + \vec{b}, \tag{43}$$

where $\vec{b}$ is a constant vector representing the same translation applied to all data points. Substituting this translation into the LLE reconstruction equation:

$$\vec{X}_i + \vec{b} \approx \sum_{j \in \mathcal{N}_i} W_{ij}(\vec{X}_j + \vec{b}). \tag{44}$$

Expanding this yields:

$$\vec{X}_i + \vec{b} \approx \sum_{j \in \mathcal{N}_i} W_{ij}\vec{X}_j + \vec{b} \sum_{j \in \mathcal{N}_i} W_{ij}. \tag{45}$$

Since the constraint $\sum_{j \in \mathcal{N}_i} W_{ij} = 1$ holds, we obtain:

$$\vec{X}_i + \vec{b} \approx \sum_{j \in \mathcal{N}_i} W_{ij}\vec{X}_j + \vec{b}. \tag{46}$$

It can be seen that the translation operation does not affect the reconstruction of the data points, and the reconstruction weights remain unchanged.

## C.2 Rotation and Scaling Invariance

Rotation and scaling are other forms of affine transformations. Suppose we apply a rotation or scaling operation to the data points:

$$\vec{X}_i' = A\vec{X}_i, \tag{47}$$

where $A$ is the rotation matrix or scaling matrix, representing the same linear transformation applied to all data points. Substituting this into the LLE reconstruction equation:

$$A\vec{X}_i = \sum_{j \in \mathcal{N}_i} W_{ij} A\vec{X}_j. \tag{48}$$

Since $A$ is a linear transformation, it can be factored out of the summation:

$$A\vec{X}_i = A \sum_{j \in \mathcal{N}_i} W_{ij}\vec{X}_j. \tag{49}$$

This indicates that rotation or scaling affects only the coordinate system, without altering the relative position relationships between data points.

In conclusion, by introducing the invariance constraint, LLE ensures that the reconstruction weights $W_{ij}$ remain unchanged when facing affine transformations such as translation, rotation, and scaling. This constraint guarantees that LLE effectively maintains the stability of the local geometric structure, preserving the relative geometric relationships between data points. As a result, LLE can accurately reflect the local structure of the data during dimensionality reduction, unaffected by coordinate transformations.

# D Explanation of Translation Degree-of-Freedom Constraint

The final step of LLE aims to find a set of low-dimensional embedding coordinates $\vec{Y}_i$ such that each $\vec{Y}_i$ can be well reconstructed from a weighted linear combination of its neighbors. This objective is achieved by minimizing the embedding cost function $\Phi(Y)$:

$$\Phi(Y) = \sum_{i=1}^{N} |\vec{Y}_i - \sum_{j\in\mathcal{N}_i} W_{ij}\vec{Y}_j|^2. \tag{50}$$

We now revisit the problem without any constraints. Assume we have found an optimal set of low-dimensional embeddings $\{\vec{Y}_1, \vec{Y}_2, \ldots, \vec{Y}_N\}$. These embeddings minimize the cost $\Phi(Y)$. If we apply a global translation to all embedding coordinates, that is, we create a new set of coordinates $\{\vec{Y}'_1, \vec{Y}'_2, \ldots, \vec{Y}'_N\}$ such that:

$$\vec{Y}'_i = \vec{Y}_i + \vec{c}, \quad i = 1, \ldots, N, \tag{51}$$

where $\vec{c}$ is an arbitrary $d$-dimensional vector. We substitute this new set of coordinates into the cost function in Eq. 50:

$$\begin{aligned}
\Phi(Y') &= \sum_{i=1}^{N} \left| \vec{Y}'_i - \sum_{j\in\mathcal{N}_i} W_{ij}\vec{Y}'_j \right|^2 \\
&= \sum_{i=1}^{N} \left| (\vec{Y}_i + \vec{c}) - \sum_{j\in\mathcal{N}_i} W_{ij}(\vec{Y}_j + \vec{c}) \right|^2 \\
&= \sum_{i=1}^{N} \left| \vec{Y}_i + \vec{c} - \sum_{j\in\mathcal{N}_i} W_{ij}\vec{Y}_j - \sum_{j\in\mathcal{N}_i} W_{ij}\vec{c} \right|^2 \\
&= \sum_{i=1}^{N} \left| \vec{Y}_i - \sum_{j\in\mathcal{N}_i} W_{ij}\vec{Y}_j + \vec{c} - \vec{c} \right|^2 \\
&= \sum_{i=1}^{N} \left| \vec{Y}_i - \sum_{j\in\mathcal{N}_i} W_{ij}\vec{Y}_j \right|^2 \\
&= \Phi(Y).
\end{aligned} \tag{52}$$

Therefore, if $\{\vec{Y}_i\}$ is a solution that minimizes $\Phi(Y)$, then any translated set $\{\vec{Y}_i + \vec{c}\}$ is also an equivalent solution with the same minimum cost. This results in an infinite number of translation-equivalent optimal solutions.

To ensure a unique and well-defined solution, LLE introduces a translation invariance constraint to eliminate this translational degree of freedom. The constraint can be formally expressed as:

$$\frac{1}{N}\sum_{i} \vec{Y}_i = \vec{0}. \tag{53}$$

Through this constraint, LLE forces the geometric center of all embedding coordinate vectors to lie at the origin of the $d$-dimensional coordinate system. This eliminates the ill-posedness of the cost function $\Phi(Y)$ under global translation, ensuring that the optimization problem admits a unique solution.

# E Explanation of Scale and Rotation Degree-of-Freedom Constraint

Building on the previous section, we now apply a global rotation to the obtained optimal embedding coordinates $\{\vec{Y}_i\}$ by introducing an orthogonal matrix $\mathbf{R}$ such that:

$$\vec{Y}'_i = \mathbf{R}\vec{Y}_i \quad i = 1, \ldots, N. \tag{54}$$

The new cost function then becomes:

$$\begin{aligned}\Phi(Y') &= \sum_{i=1}^{N} \left| \mathbf{R}\vec{Y}_i' - \sum_{j \in \mathcal{N}_i} W_{ij} \mathbf{R}\vec{Y}_j' \right|^2 \\ &= \sum_{i=1}^{N} \left| \mathbf{R}(\vec{Y}_i' - \sum_{j \in \mathcal{N}_i} W_{ij} \vec{Y}_j') \right|^2 .\end{aligned} \tag{55}$$

From the definition of the Euclidean distance, we have:

$$\begin{aligned}|\mathbf{R}\vec{v}|^2 &= (\mathbf{R}\vec{v})^T(\mathbf{R}\vec{v}) \\ &= \vec{v}^T\mathbf{R}^T\mathbf{R}\vec{v} \\ &= \vec{v}^T\mathbf{I}\vec{v} \\ &= \vec{v}^T\vec{v} \\ &= |\vec{v}|^2.\end{aligned} \tag{56}$$

Combining Eq. 55 and Eq. 56 yields that:

$$\Phi(Y') = \sum_{i=1}^{N} \left| \vec{Y}_i' - \sum_{j \in \mathcal{N}_i} W_{ij} \vec{Y}_j' \right|^2 = \Phi(Y). \tag{57}$$

This means that if $\vec{Y}_i$ is an optimal solution, any rotated set $\mathbf{R}\vec{Y}_i$ is also an optimal solution. This results in infinitely many rotation-equivalent optima, making the problem ill-posed.

Similarly, if we apply a global uniform scaling to all coordinates by a scalar $\alpha$ such that:

$$\vec{Y}_i' = \alpha\vec{Y}_i. \tag{58}$$

Substituting this into the cost function yields:

$$\begin{aligned}\Phi(Y') &= \sum_{i=1}^{N} \left| \alpha\vec{Y}_i' - \sum_{j \in \mathcal{N}_i} W_{ij} \alpha\vec{Y}_j' \right|^2 \\ &= \sum_{i=1}^{N} \left| \alpha(\vec{Y}_i' - \sum_{j \in \mathcal{N}_i} W_{ij} \vec{Y}_j') \right|^2 \\ &= \sum_{i=1}^{N} \alpha^2 \left| \vec{Y}_i' - \sum_{j \in \mathcal{N}_i} W_{ij} \vec{Y}_j' \right|^2 \\ &= \alpha^2\Phi(Y).\end{aligned} \tag{59}$$

To minimize the cost function, the solution would always tend toward $\alpha \to 0$, causing all embedding coordinates to collapse to the origin and lose their meaning. Therefore, it is necessary to fix the scale of the embedding.

To address these issues, LLE introduces scaling and rotation invariance constraints. It enforces the sample covariance matrix of the embedding coordinates to be equal to the $d \times d$ identity matrix. The constraint can be formally written as:

$$\frac{1}{N}\sum_i \vec{Y}_i\vec{Y}_i^T = \mathbf{I}. \tag{60}$$

This constraint forces the off-diagonal elements of the covariance matrix to be zero, indicating that different coordinate dimensions in the embedding space are uncorrelated. This simplifies the interpretation of the embedding and contributes to the uniqueness of the solution. At the same time, enforcing ones on the main diagonal ensures that each dimension in the embedding space has equal variance of one. This keeps the reconstruction error measured on each dimension at the same scale and prevents certain dimensions from being unnecessarily magnified or compressed. As a result, the low-dimensional embedding obtained by LLE attains a unique geometric structure with a standardized scale.